Dragon Genetics Worksheet #1: "I'm All Keyed Up"

Name: _____

Date: _____

Class/Period: _____

INSTRUCTIONS

Refer to the key below, and answer the following questions.

Dragon Key

Upper case letters represent dominant alleles; lower case letters represent recessive alleles

N = LONG NECK n = short neck	E = Red Eye e = white eye
H = HORN PRESENT h = horn absent	F = FIRE BREATHING f = not fire breathing
G = GREEN BODY g = gray body	L = LONG TAIL l = short tail
S = SPIKES ON END OF TAIL s = no spikes on end of tail	R = RED WINGS r = yellow wings
T = THREE TOES t = four toes	W = YELLOW BELLY w = white belly
B = BLACK TAIL SPIKES b = red tail spikes	K = FRECKLES k = no freckles
X = X chromosome / no ear frills (XX = female)	Y = Y chromosome / ear frills present (XY = male)

1. What letters are used to represent wing color? _____
2. What letters are used to represent neck length? _____
3. The letter "Y" is used as a symbol to represent _____.
4. The letter "e" is used to represent _____.

An upper case, or capital, letter is used to represent a dominant trait. A lower case, or small, letter is used to represent a recessive trait. Dominant traits completely mask and/or suppress recessive traits.

Refer to the key, and answer the following questions.

5. List six dominant traits shown in the key.

_____ _____

_____ _____

_____ _____

6. List six recessive traits shown in the key.

_____ _____

_____ _____

_____ _____

Dragon Genetics Worksheet #2: "Would Someone Please Take Charge?"

Name: _____

_____

Date: _____

Class/Period: _____

_____

EXPLANATION

The simplest form of genetic inheritance for a single trait involves receiving one piece of genetic information (one allele) from the mother and one piece of genetic information (one allele) from the father. Sometimes the information from the mom and dad is identical, and sometimes it is not. The resulting combination of alleles is referred to as a genotype; the physical appearance or expressed trait that is coded for is referred to as the phenotype. For example, the genotype for our long necked dragon could be either NN or Nn; the phenotype for this trait would be a long neck.

INSTRUCTIONS

Using the key from Worksheet #1 as a reference, fill in the missing symbols for alleles (genotypes) or descriptions of physical appearance (phenotypes) in the table below. The first two rows are done for you. Remember, dominant traits appear when either one or two copies of the dominant allele are present. For example, "NN" and "Nn" both cause the dragon's neck to be long. Only in the case of "nn" would the dragon's neck be short because no dominant allele for a long neck is present.

Trait	Symbols for Alleles	Appearance
Neck size	NN or Nn	Long neck
Neck size	nn	Short neck
Spikes on tail?		No spikes
Eye color	EE	
Horn? (present or absent)	hh	
Fire breathing?	_____ or _____	Breathes fire
Belly color	WW or Ww	
Color of spikes		Red
Tail length	_____ or _____	Long

Dragon Genetics Worksheet #4: "Enter the Dragon"

Name: _____

_____

Date: _____

Class/Period: _____

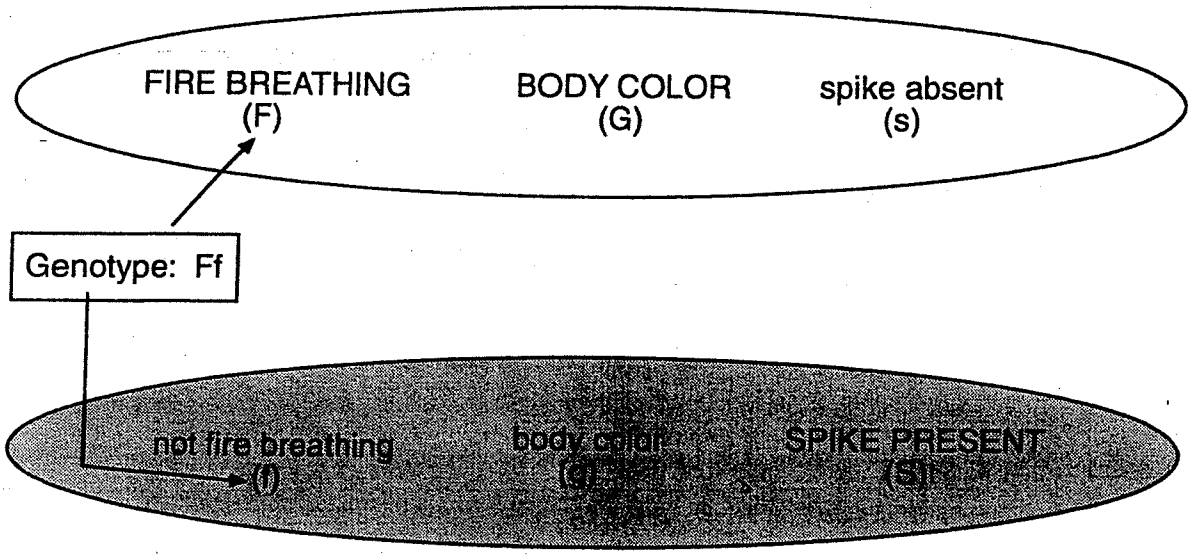
_____

INSTRUCTIONS

Traits are encoded on structures called chromosomes. When an egg is fertilized by a sperm, the resulting offspring should receive an equal number of chromosomes from the mom and from the dad.

A pair of hypothetical dragon chromosomes is shown below. In this example, the white chromosome is from the mom and the gray chromosome is from the dad. Determine the genotype and phenotype that would result in the offspring of these two dragons, and fill in the missing information in the data table below. (The first line is filled in for you.)

Pair of Dragon Chromosomes



Data Table

TRAIT	GENOTYPE	PHENOTYPE
Fire breathing or not	Ff	Fire breathing
Body color		
Spikes (present or absent)		

Dragon Genetics Activity #5

OVERVIEW

In this activity, you will "mate" with a dragon of the opposite gender to create a "baby dragon." You will then use your knowledge of genetics to determine the genotype and phenotype for each of 13 traits in your baby dragon. You will also construct a picture of your baby dragon, by cutting out the appropriate pieces from the blackline drawing on page S8, taping/gluing each piece in the proper place, and coloring each piece in the appropriate color.

MATERIALS NEEDED

1 complete set of 4 dragon chromosomes, provided in either an "egg" or a "sperm"
Dragon Genetics Activity #5 Worksheet (page S7)
Dragon blackline drawing (page S8)
Colored pencils (red, white, green, gray, orange, yellow, black)
Scissors
Tape or glue

PROCEDURE

1. Your teacher should have given you either a dragon "egg" or a dragon "sperm" with a number on it. Find the other person in your class who has the "egg" or "sperm" with the matching number. This person is now your "mate."

2. Remove all of the chromosomes from your respective "egg" and "sperm." (All four chromosomes in the egg should be pink; all four chromosomes in the sperm should be light blue.)

3. Find chromosome #1 (C#1) from both the mother and father. (The chromosome number is located at the bottom, center of each chromosome.) Pair these two chromosomes together.

4. Repeat step 3 for each of the remaining chromosomes (that is, pair female chromosome #2 with male chromosome #2; pair female chromosome #3 with male chromosome #3; female chromosome #4 with male chromosome #4.) At the end of this step you should have all eight chromosomes (four pairs) that make up your "baby dragon."

5. Using the key at right, determine your "baby dragon's" genotype and phenotype for all of the traits listed on the chromosomes. Record this information in the proper spaces of the data table on page S7.

6. Refer to phenotypes you listed on the data table. For each phenotype listed, cut out the corresponding dragon body part found on the blackline drawing on page S8.

7. Using tape or glue, assemble the blackline pieces you cut out, to create your "baby dragon." Be sure to color your "baby dragon's" features in the appropriate colors, according to the phenotypes you have listed.

Dragon Key

Upper case letters represent dominant alleles; lower case letters represent recessive alleles

N = LONG NECK n = short neck	E = Red Eye e = white eye
H = HORN PRESENT h = horn absent	F = FIRE BREATHING f = not fire breathing
G = GREEN BODY g = gray body	L = LONG TAIL l = short tail
S = SPIKES ON END OF TAIL s = no spikes on end of tail	R = RED WINGS r = yellow wings
T = THREE TOES t = four toes	W = YELLOW BELLY w = white belly
B = BLACK TAIL SPIKES b = red tail spikes	K = FRECKLES k = no freckles
X = X chromosome / no ear frills (XX = female)	Y = Y chromosome / ear frills present (XY = male)

8. At the end of this activity, place all four pink chromosomes back into the "egg" and all four blue chromosomes back into the "sperm."

Dragon Genetics Worksheet #5:

Name: _____

Date: _____

Class/Period: _____

INSTRUCTIONS

Using the key on page S6, determine your "baby dragon's" genotype and phenotype for all of the traits listed on the chromosomes. Also determine whether your "baby dragon" is heterozygous or homozygous for each trait. Record all of this information in the proper spaces of the data table below.

Data Table

Chromosome pair number	Trait	Genotype	Homozygous/ Heterozygous	Phenotype
1	Neck length			
	Eye color			
	Horn?			
	Spikes?			

Chromosome pair number	Trait	Genotype	Homozygous/ Heterozygous	Phenotype
2	Tail length			
	Body color			
	Color of wings			
	Number of toes			

Chromosome pair number	Trait	Genotype	Homozygous/ Heterozygous	Phenotype
3	Belly color			
	Color of spikes			
	Freckles?			

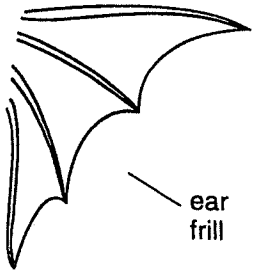
Chromosome pair number	Trait	Genotype	Homozygous/ Heterozygous	Phenotype
4	Fire breathing?			
	Ear frills?			

DRAGON BLACKLINE MASTER

head with freckles

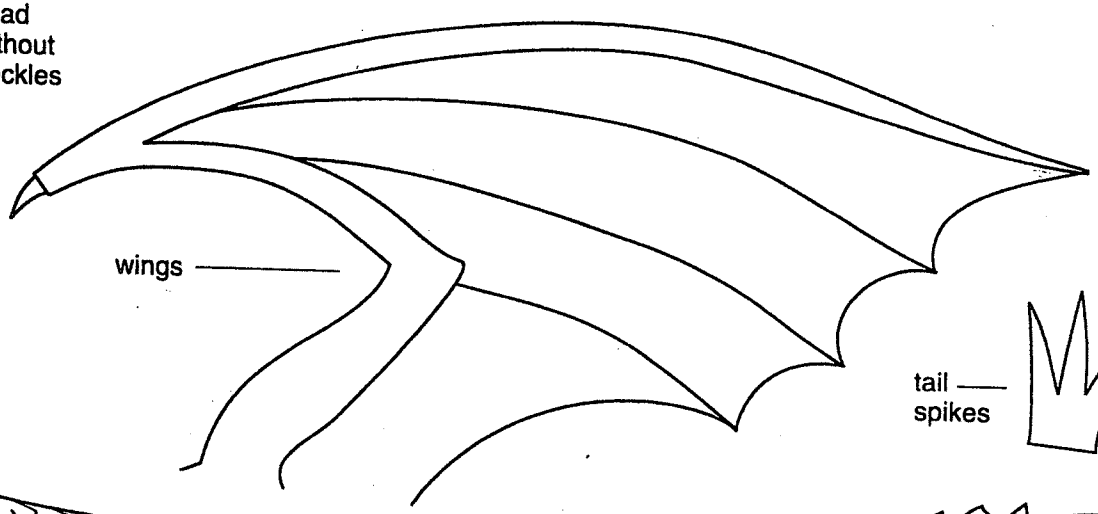
head without freckles

fire breath

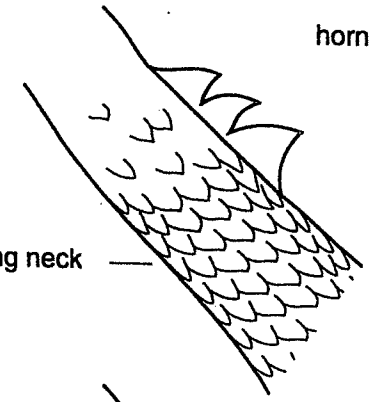


ear frill

horn

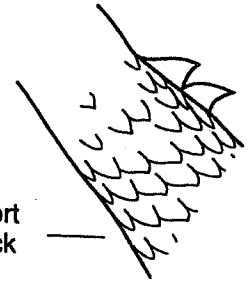


wings

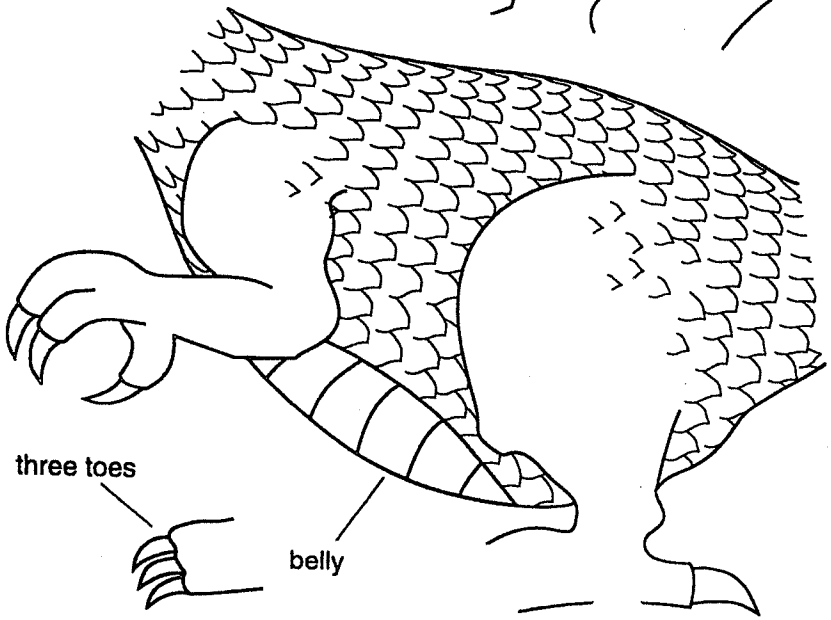


long neck

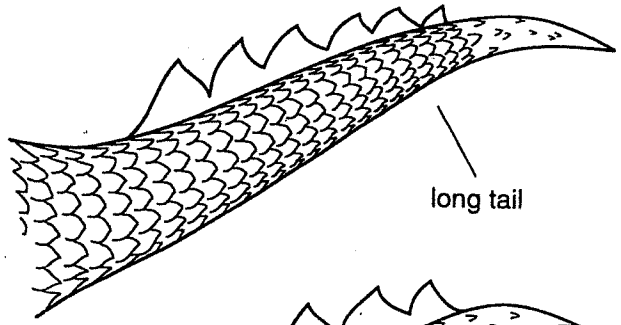
tail spikes



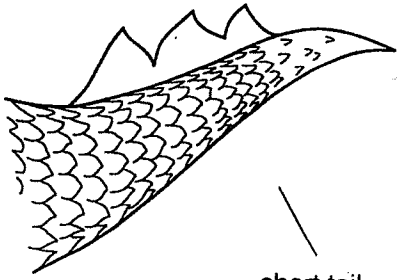
short neck



belly



long tail



short tail

three toes

four toes